



797F Trucks Brake Accumulators Removal & Install Tool

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

Title 797F Truck Brake Accumulators R&I..... **Error!**
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1.0 Introduction

Safe, efficient and quality maintenance and repair work is important to any mine in reaching production goals. When dealing with large size equipment such as the Caterpillar 797F trucks, the maintenance activities usually include tasks that require the handling of heavy and large components.

To achieve the objective of safe, efficient and quality repairs, the maintenance department has to ensure that the personnel is properly equipped with the necessary training, support information (technical and processes information), infrastructure, tools and support equipment.

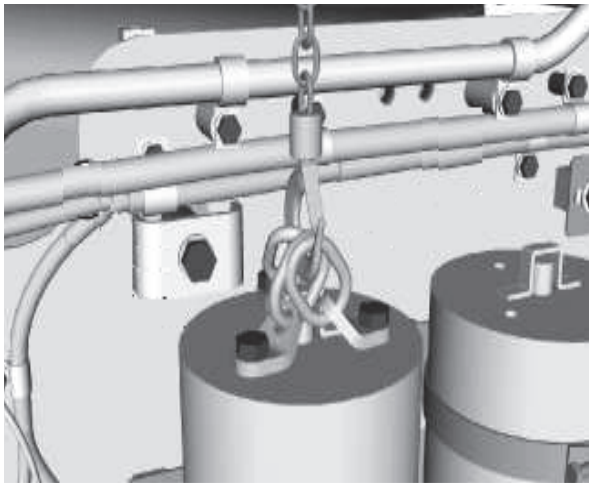
This Best Practice refers to the design, use, and benefits of a special tool developed for the removal and installation of brake accumulators of 797F trucks in order to contribute to the objectives mentioned above.

2.0 Best Practice Description

2.1 Opportunity Detection

Throughout the analysis of the removal and installation task of the 797F truck brake accumulator, 4 large areas of improvement opportunity were detected. These opportunities motivated the development of the tool proposed in this Best Practice.

Opportunity areas



Safety

The current procedure recommended to replace brake accumulators (Mean # KENR8382-08), involves the use of chains / straps, lifting tackle(s), and adequate lifting equipment. Due to the sensitive nature of the procedure, successful application relies on the experience of the technician(s) performing the task. The technician(s) must maintain control and balance of the brake accumulator at all times to avoid the risk of damage to the part or equipment from striking the part during movement.

Working with suspended loads is a hazardous task on its own. (Weight of the brake accumulator: 69 Kg).

Resources (Labor)

Normally, 3 technicians are assigned to the performance of this task, and 1 operating the overhead crane. This means that 4 people are needed to fulfil the task following the recommended procedure.

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With the use of the proposed tool, one of the technicians can be released, reducing the required manpower for the task.

Quality

The observations made to the replacement of brake accumulators, also found some substandard events regarding quality, which were mainly caused by the lack of control of the suspended load.

2.2 Improvement solution proposed

The solution proposed is the use of a tool specially designed for this application (Image 1). It uses an eyebolt, together with lifting equipment (Overhead crane) (Image 2), and the tool that we will call ARIAF-DMH (for its Spanish acronym: Accessory for Removal and Installation of Brake Accumulators – Ministro Hales Division)

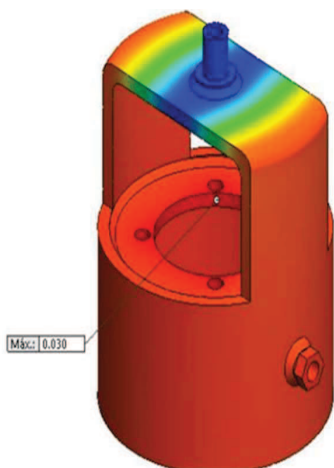


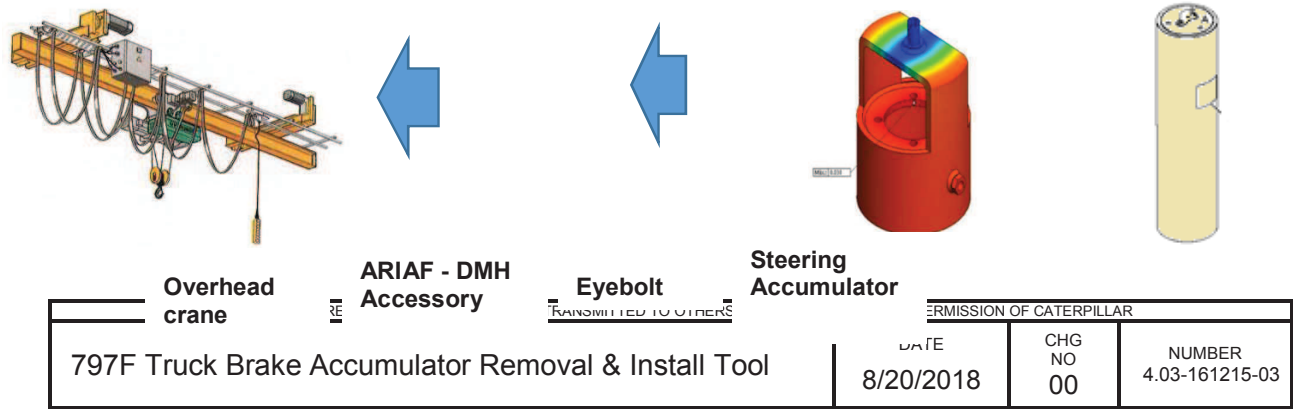
Image 1

Accessory for the Replacement and Installation of Brake Accumulators



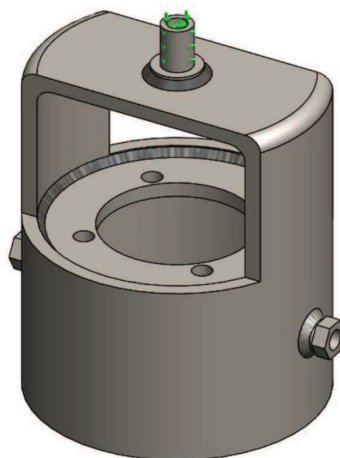
Image 2

Eyebolt Installation in the ARIAF-DMH



The combination of a stacker and a rigid accessory gives us the necessary safety and maneuverability to perform this task with significantly improved control. This solution changes the original work situation with a suspended load to a supported load, which is much more favorable with regard to maneuvering safely.

The following pictures show the ARIAF-DMH equipment operating in our workshops.



3.0 Steps for implementation

The general steps we recommend following to implement this Best Practice are:

- 3.1 ARIAF-DMH manufacture
- 3.2 Eyebolt selection and acquisition
- 3.3 ARIAF-DMH Technical certification
- 3.4 Practical tests on accumulator replacement tasks
- 3.5 Final adjustments
- 3.6 SOP (Standard Operation Procedure) documentation
- 3.7 Staff training
- 3.8 Start-up and benefit validation

3.1 ARIAF-DMH manufacture

The ARIAF-DMH was manufactured locally according to the following design specifications. This design was made locally in the Finning – DMH contract.

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3.2 Selection and acquisition of the recommended eyebolt



Image 3

½ Inch Eyebolt

3.3 Technical Certification of the brake accumulator R&I tool (Welding and Capacity)

Welding certification and capacity analysis of the tool once it is manufactured are important.

08415-ACUMULADOR DE FRENO

MEMORIA DE CALCULO PARA HERRAMIENTA DE ACUMULADOR DE FRENO.

REV	DESCRIPCION	INGENIERIA			APROBACION	
		Por	Jefe proy.	Fecha	Jefe de proy.	Fecha
0	GR Ingeniería.	Giovani Rojas	Giovani Rojas	27-11-15		
B	Revisión Interna	Ricardo Vega V.	Ricardo Vega V.	27-11-15		
A	Revisión Interna	Marcelo Vargas A.	Marcelo Vargas A.	27-11-15		

Report on the Structural Calculus and Quality Certification of the ARIAF-DMH

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Figure 7.7. Displacement on the element (0.030 mm).

Testing examples / Stress analysis
done on the ARIAF-DMH

3.4 Practical tests on a brake accumulator R&I task

Once the accessory is received, we recommend running operational tests in the workshop, in a safe and controlled environment, in order to identify and make the final adjustments required.

Final adjustments

Adjust what is detected and required.

3.6 SOP (Standard Operation Procedure) documentation

To ensure the right operation of this accessory, we recommend documenting the standard procedure of use (SOP).

We also recommend checking safety standards specific to your operation in order to adapt the SOP to your specific needs.

3.7 Staff training

Adequate training is important for the staff who are going to use the ARIAF-DMH habitually on accumulator replacement tasks.

We recommend including in this training both the standard procedure (SOP) review and practice in a safe and controlled environment.

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3.8 Start-up and benefit validation

Once the ARIAF-DMH is operating, there must be a follow-up of proper use and benefit records.

Once the expected benefits are obtained, they must be improved / modified with the particular Standard Jobs Planning and Programming Area, especially the new machine shutdown and man hour times.

4.0 Benefits

According to our observations and records of using this accessory in our operations, the benefits registered so far show the following improvements.

	Hazard level without ARIAF - DMH	Hazard level with ARIAF - DMH	Number of technicians without ARIAF - DMH	Number of technicians with ARIAF - DMH
Replacement of steering accumulator	Medium	Low	3	2

The main benefits are located in the areas of: Safety, Manpower Use, and Execution Quality.

Safety: Suspended load mode is replaced by supported load, reducing hazard levels significantly.

Man hours (MH used): Releasing one person from supporting this task means a 33% decrease in MH, which can be assigned to other works.

The decrease in the MH invested in this task will have a positive influence on the MR (Maintenance Ratio) results of the fleet.

Quality: Although it is hard to quantify, we have detected improvements on the quality of the steering accumulator R&I operation. The control on the maneuvers and effort of the staff has a positive effect on the quality of the work done.

Improvements in quality will have a positive influence on the MTBS results of the fleet.

5.0 Resources required

The resources required are those specified in this Best Practice.

6.0 Additional support / References

The following information is attached to this Best Practice

- Manufacture Scheme / Drawing

Note: The Report on structural calculus and certification may be provided upon request.

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7.0 Other Best Practices related

N/A

8.0 Acknowledgements

Best Practice generation, implementation, and documentation

Fernando Valdés
Contract Manager
Ministro Hales Contract (Codelco Chile)
Phone: +56 984180500
E-mail: fernando.valdes@finning.com

Marcos Palma Albornoz
CI department Manager
Ministro Hales Contract (Codelco Chile)
Phone: +56 961254692
E-mail: marcos.palma@finning.com

Best Practice written by:

Juan Crooker Acosta/ Juan Carlos Concha
Best Practice Program Coordinator Finning
Phone: +56 966163251
E-mail: juan.crooker@finning.com

Abelardo A. Flores
Market Professional - MEM
Caterpillar Global Mining

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